

Applied A.I. Solutions

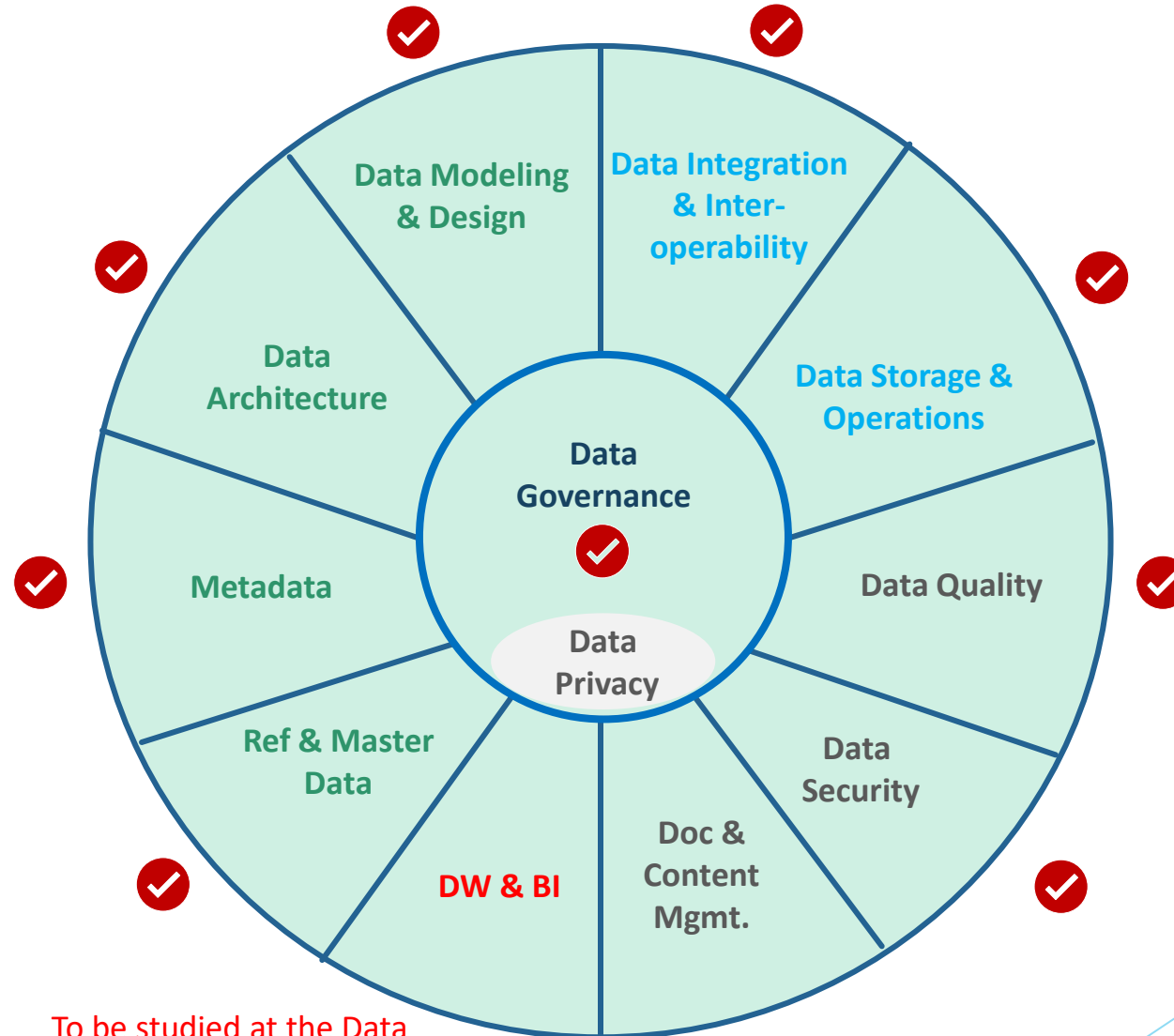
Foundations of Data Management

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DATA SECURITY

The DAMA Wheel

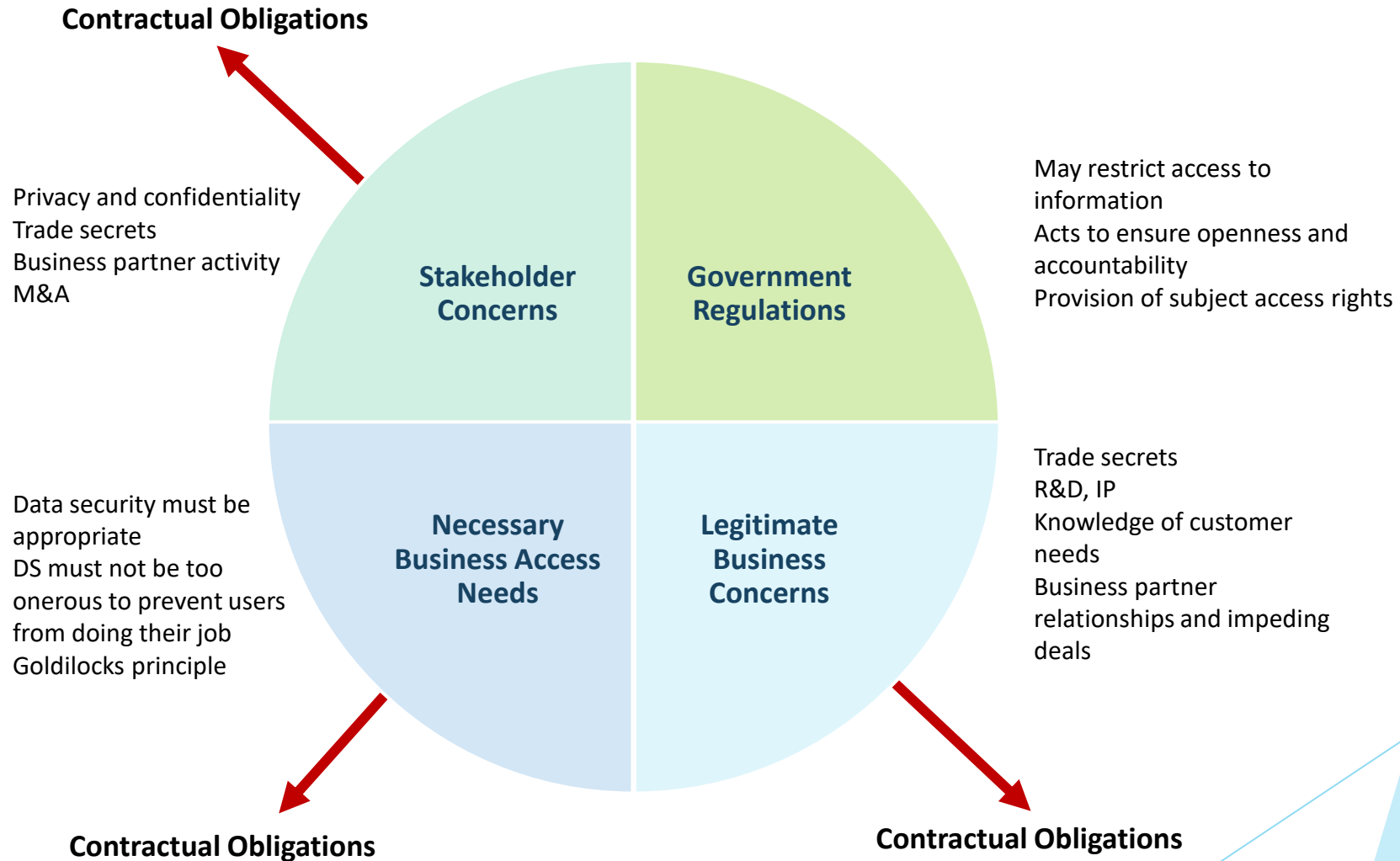


To be studied at the Data Visualization Techniques course

1. INTRODUCTION

- **Protect information assets** in alignment with privacy and confidentiality regulations, contractual agreements, and business requirements
- Ensure that the right people can use, update data in the right way, and that all inappropriate **access**, update is restricted

Requirements come from:



Data Security Management Framework

Definition

Definition, planning, development, and execution of **security policies and procedures to provide proper authentication, authorization access, and auditing of data and information assets**

DS programs are guided by the following principles:

1. Collaboration
2. Enterprise approach
3. Proactive management
4. Clear accountability
5. Metadata-driven
6. Reduce risk by reducing exposure

Drivers

1. Risk Reduction

- Identify and classify sensitive data assets
- Determine how each asset needs to be protected
- Identify how the information interacts with business processes (access)

2. Business Growth

- Security as an Asset

Goals

1. **Enabling** appropriate, and **preventing** inappropriate **access** to data assets
2. **Enabling compliance** with **regulations**, policies for privacy, protection, and confidentiality
3. Ensuring that **stakeholder requirements** for privacy and confidentiality are met



Business Drivers

Inputs

- Business strategy
- Business rules
- Regulatory requirements
- Enterprise architecture
- Enterprise data model



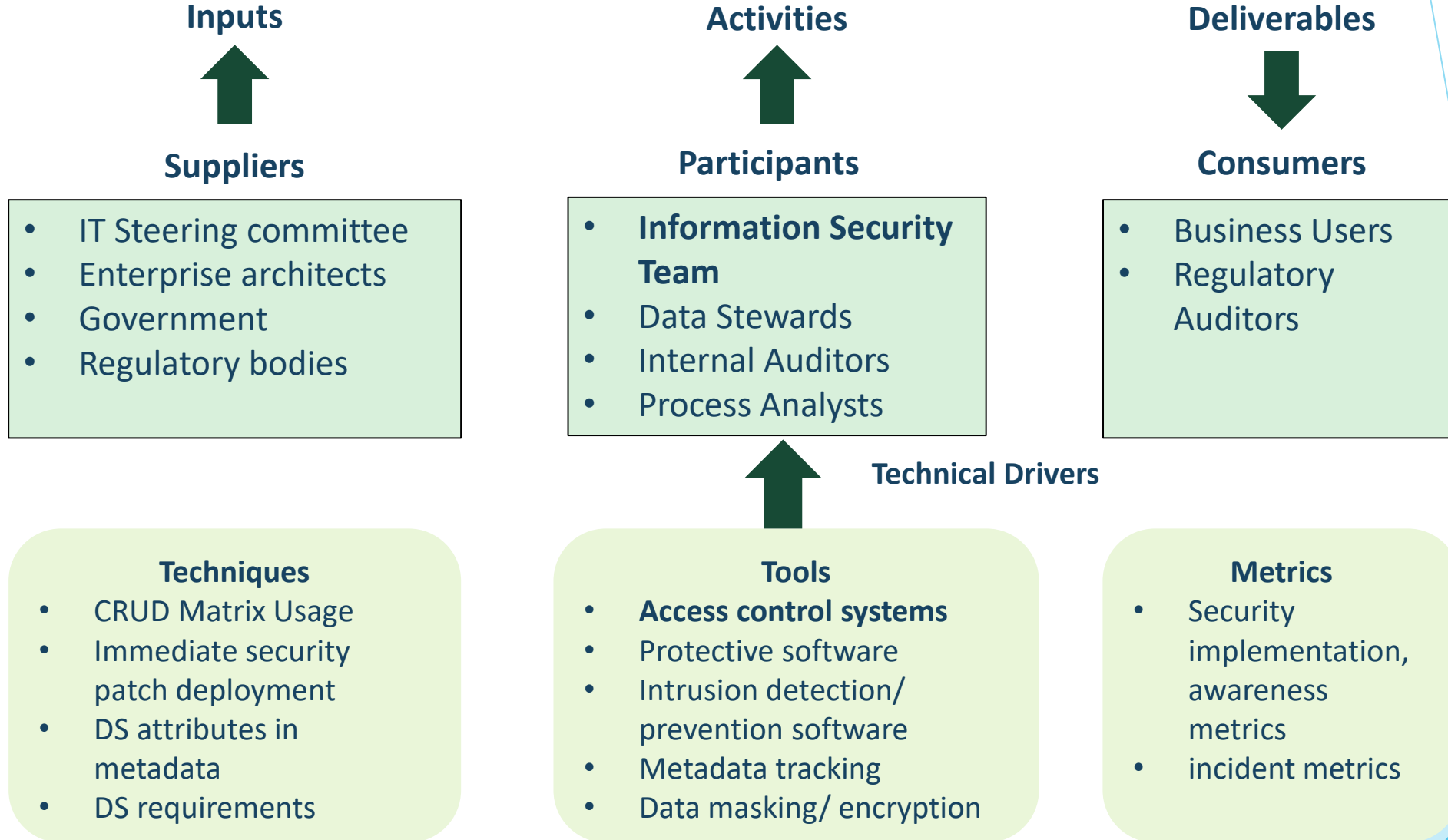
Activities

1. Identify data security requirements
2. Define data security policy and standards
3. Assess current security risks
4. Implement controls and procedures



Deliverables

- Data privacy and confidentiality standards
- D Sec access controls
- Regulatory compliant
- Documented security classifications



Data Security Essential Concepts

- Risks - Risk Classification
- Security Processes
- Masking / Encryption
- Vulnerability
- Threats, Hacking / Phishing / Malware
- Data Integrity
- Security Processes

- **Risks**

- Main Risk Factors**

- Damage on revenue, business operations and reputation
 - Cost to prevent the threat, remediate of vulnerabilities

- Risk Classification**

- Critical – Personal Information / financial / health value
 - High – potential direct PI / financial / health value
 - Moderate – little tangible value

- **Security Processes**

1. **The Four A's**

- Access
- Authentication
- Authorization
- Audit

A light blue bracket on the left side of the slide groups the four items under 'The Four A's'. To the right of the bracket, the word 'Entitlement' is written in a bold, dark blue font.

Entitlement

2. **Monitoring**

- **Masking**

Data is made less available without losing the meaning of the data or relationships to other datasets

Types

- **Persistent Data Masking**
 - **In-flight persistent masking**
(source $\neq$ destination)
 - **In-place persistent masking**
(source = destination)
- **Dynamic Data Masking (DDM)**

Methods

- **Substitution**
- **Encryption**
- Shuffling
- Value variance
- Randomization
- Expression masking
- Key masking

Types of Data Security Restrictions

- **Confidential Level**

- Confidential Data
 - For general audience
 - Internal use only
 - Confidential
 - Restricted confidential
 - Registered confidential

- **Regulations**

- Regulated Data
 - Personal Identification Inf. / PIPEDA
 - Financial Sensitive Data
 - Medically Sensitive Data / PHIPA
 - Educational Records

Important: Confidentiality does not imply regulatory compliance

